

FCT SERVICE NOTE

SN05082410a

Class A Service information enclosed.

Date: August 24, 2005

Subject: LightSpeed RT CT Number Fix Patch

Effectivity: LightSpeed RT with 05MW14.5 software or earlier.

NOTE: The Install of this Patch is Mandatory to meet Chinese regulatory standards (YY-0310).

Details:

This LightSpeed RT CT Number fix for air patch is **ONLY** applicable for **LightSpeed RT** systems installed in China and all production **LightSpeed RT** systems to be installed in China.

NOTE: This patch CD should be run every time after a detailed calibration (including Auto CT Number Adjust) is successfully completed on LightSpeed RT systems.

Patch Loading Procedure:

Feature Summary

This patch is used to fix the air CT number shift in LightSpeed RT systems. According to Chinese regulatory standards (YY-0310), the air CT number for CT scanners sold in China shall meet the specification of -1000 ± 10 HU. Most of RT systems either in China installed base (IB) or currently manufactured may not meet this standard.

To fix the problem, the patch rescales pcal vectors and adjusts beam hardening coefficients for each kv, slice mode, bowtie filter and focal spot in the calibration database. Air CT number computed using these rescaled vectors will meet the specification of -1000 ± 10 HU.

The patch must be run every time a detailed calibration (including Auto CT Number Adjust) is done. However, the pcal vectors can only be rescaled once between two consecutive detailed calibrations. To prevent multiple rescaling, a "touch" file is generated in /usr/g/cal_db when this patch is run. During the next installation, the system will compare the time stamp of the calibration vectors to the time stamp of the "touch" file, to decide whether the patch should be run --- only if the time stamp of cal vectors is newer than the time stamp of the "touch" file, the rescale is allowed.

Patch Loading Instructions:

1. Before loading this patch, please make sure the software is ensure **(05MW14.5)** software release Q2 2005 Software Set is installed. Otherwise, load the 05MW14.5 before installing this patch.
2. For manufacturing, if the detailed calibration is completed and system states have been saved, please skip this step and go to step 3. For installed base, if this is the first time loading the Patch, please perform the following detailed calibration procedures:
 - a. Z-Slope calibration
 - b. DAS Gain calibration
 - c. Collimator calibration
 - d. Detailed calibration
 - e. Auto CT number adjust
3. Save the system states into System States DVD.
4. Shutdown the application: **Service Desktop -> Utilities -> Application Shutdown**
5. Open the front cover of the console so that the **HOST** computer is accessible.
6. Insert “RT CT Numberfix” patch CD into DVD drive in the **HOST** computer.
7. Open a Unix Shell.
8. In the Open Unix Shell, login as a super user:
 - a. Type: **su – <Enter>**
 - b. Type super user’s password **<Enter>**
9. After login as a super user, at the same Unix Shell,
 - a. Type: **mount /mnt/cdrom <Enter>**
 - b. Type: **cd /mnt/cdrom <Enter>**
 - c. Type: **./install_patch -p RT_CTNum_Fix-1-1.i386.rpm <Enter>**
10. It takes about 15 minutes for the installation to be completed.
11. In the Open Unix Shell
 - a. Type: **./uninstall_patch <Enter>**
 - i. Select: the number associated with this patch “RT_CTNum_Fix-1-1”
 - b. Type: **cd /usr/g/ctuser <Enter>**
 - c. Type: **umount /mnt/cdrom <Enter>**
 - d. Type: **exit <Enter>**
12. Eject the Patch CD.
13. In the Open Unix Shell:

- a. Type: *st* <Enter>
14. Once the application is reloaded, perform an air-only scan using the following protocol, and then measure the air CT number:
 - a. 120kV, 100mA, 1.0s, 4x1.25mm/4i, large SFOV, 50cm DFOV, standard filter
 - b. Measure air CT number using a default size circular ROI at the center of each image.
 - c. Air CT number should be within: -1000 +/- 10
 15. Scan water section (QA3) of a centered 20cm QA phantom and measure the water CT number:
 - a. 120kV, 260mA, 1.0s, 4x1.25mm/4i, small SFOV, 25cm DFOV, standard filter
 - b. Measure water CT number using a default size circular ROI at the center of each image.
 - c. Water CT number should be within: 0 +/- 5.
 16. If either one of above two scans fails the specified CT number range or the scan “resume” due to calibration file error, restore the system states from System States DVD, and the system should go back to the state prior to the patch installation.
 17. If both scans pass the specified range of CT Number for air and water, save the systems states into System States DVD.

Resolution: N/A

Appendix: - Orderable Parts:

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| ☐ 05MW14.5 Software Collector | 5134312 |
| ☐ 05MW14.5 LFC | 5134798-200 |

James Lehouillier
Console Service Methods Design Engineer